Operating Instructions

Diesel engine

12 V 4000 Sx3

16 V 4000 Sx3

MS150028/00E

Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
12V4000S23	118.7 kW/cyl.	4A, continuous operation, unrestricted
16V4000S23	116.6 kW/cyl.	4A, continuous operation, unrestricted

Table 1: Applicability

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1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:
- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

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1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.
- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

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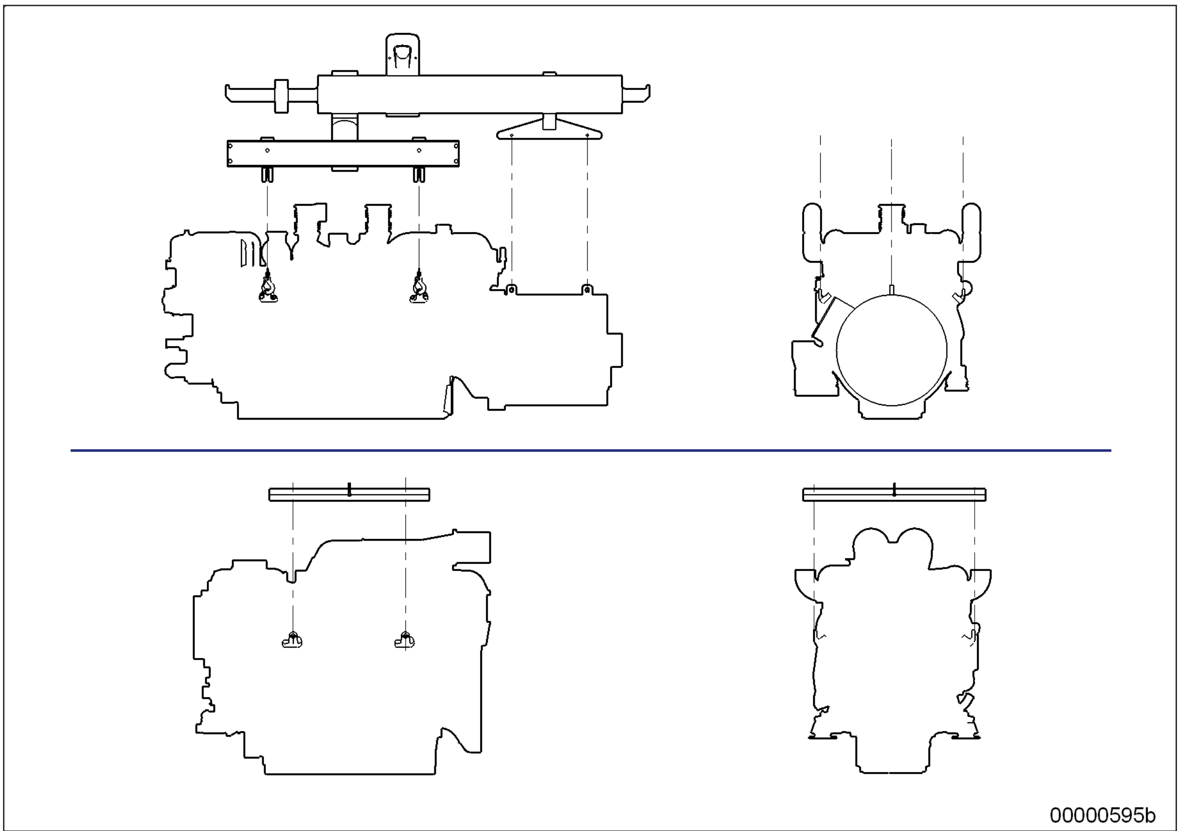
1.3 Transport

Transport



Always use the lifting eyes on the engine and generator/gearbox when transporting gensets.

Always use the lifting eyes on the engine when transporting an engine separately.

Only use transport and lifting devices approved by MTU.

The engine/genset must only be transported in installation position: maximum permissible diagonal pull 10°.

Remove any loose parts on the genset.

Raise the engine/genset slowly. The lifting ropes or chains must not make contact with the engine or its components. Readjust lifting device as necessary.

Pay attention to the center of gravity of the engine/genset: Refer to installation drawing.

For special packaging with aluminium foil: Suspend the engine/genset by the lifting eyes on the bearing pedestal or transport by means of handling equipment (forklift truck) capable of bearing the load.

Fit the crankshaft transportation lock on the engine and fit the engine mount locking devices prior to transport.

Secure the engine/genset such as to preclude tipping during transport. Secure such as to preclude slipping and tipping when driving up or down inclines and ramps.

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Placement after transport

Place the engine/genset on a firm, flat surface only.

Make sure that the consistency and load-bearing capacity of the ground or support surface is adequate.

Never set an engine down on the oil pan unless expressively authorized to do so by MTU on a case-to-case basis.

1.4 Crankshaft transport locking device

Special tools, Material, Spare parts

Designation / Use	Part No.	Qty.
Torque wrench, 10-60 Nm	F30452769	1
Torque wrench, 60-320 Nm	F30452768	1
Engine oil		

Transport locking device

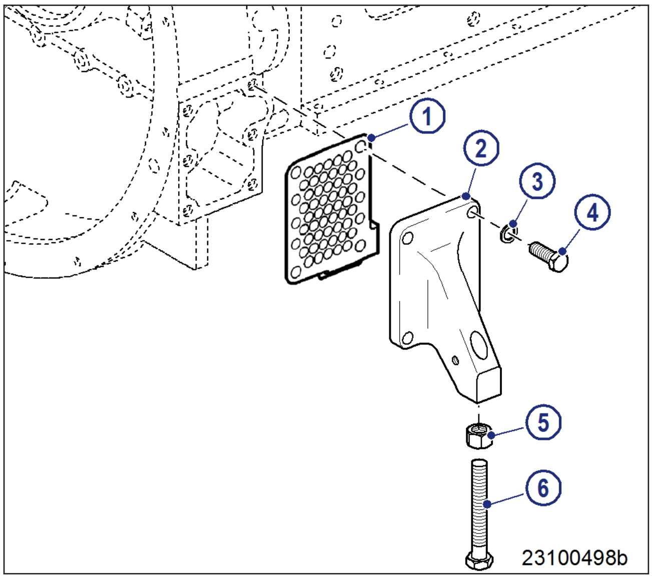
Note: The transport locking device on both sides protects the crankshaft bearings from impact and vibration damage during engine transport.

For installation and removal of the transport locking device, follow the instructions below:

1. The transport locking device must remain installed on both engine sides as long as possible during engine installation in order to avoid damage.
2. Prior to every engine transport, the transport locking device must be reinstalled on both engine sides according to the instructions.
3. If the engine is to be moved together with the generator, the transport locking device for the generator must also be installed.
4. Always use the screws supplied with or installed in the transport locking device to secure it on the engine.
5. Starting or barring the engine is allowed only with the transport locking device removed. If the generator is already installed on the engine, ensure that the transport locking device of the generator is also removed.

Removing guard plates and engine supports (if applicable) on driving end (KS)

1. Remove screws (4) om both sides and take off with washers (3), guard plates (1) and engine supports (2).
2. Store the removed parts of the transport locking device carefully for any reuse.



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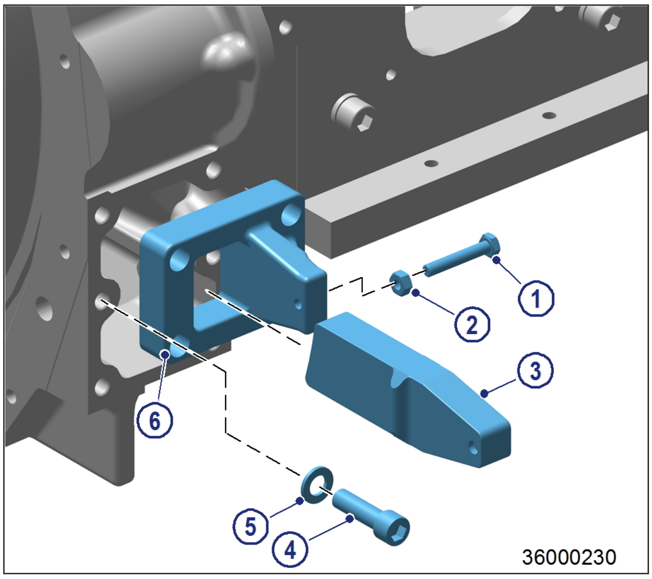
Fitting the transport locking device on the driving end (KS)



Note: Screw plate (6) onto the upper part of the openings only.

1. Fasten the plates (6) with screws (4) and washers (5) at the lateral openings on both sides of the flywheel housing and tighten to the specified tightening torque.

Name	Size	Type	Lubricant	Value/Standard
Screw	M16	Tightening torque	(Engine oil)	250 Nm +25 Nm

2. Fit locknut (2) on screws (1) up to the end of the thread.
3. The long side of retainer (3) must point downwards. Insert retainer (3) through the openings of plate (6).

Note: Retainer (3) must tension only the flywheel, not the ring gear.

4. Insert screw (1) in bores of retainer (3) until retainer (3) is locked in position.

Note: Screw (1) must be tightened alternately on both sides of the flywheel housing.

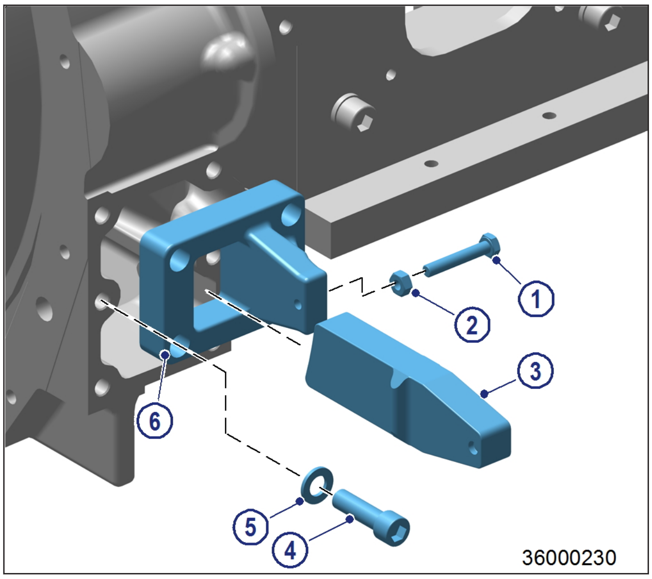
5. Tighten screw (1) to specified tightening torque .

Name	Size	Type	Lubricant	Value/Standard
Screw	M10	Tightening torque	(Engine oil)	30 Nm +3 Nm

6. Fit locknut (2) of screw (1) on plate (6) and tighten.

Removing the transport locking device from the driving end (KS)

1. Release the locknuts (2) on both sides of the flywheel housing, remove screw (1) and take off retainer (3).
2. Remove screws (4) and take off with washers (5) and plates (6).
3. Store the removed parts required for re-installation carefully together with this documentation.



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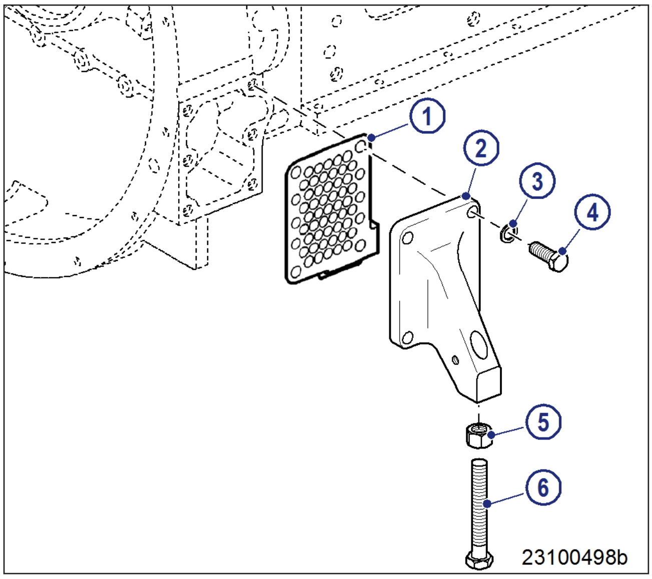
Installing guard plates and engine supports (if applicable) on driving end (KS)

Note: Always use the screws supplied with or removed from the guard plates and engine supports to secure them on the engine.

1. Install engine supports (2) on both sides with guard plates (1) washers (3), and screws (4).
2. Tighten screws (4).



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